

DSC 411/511
Advance Data Mining Algorithms and Applications

Fall 2026

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Location: MSB 104



Lecture 03
Data Preprocessing

Previous class:

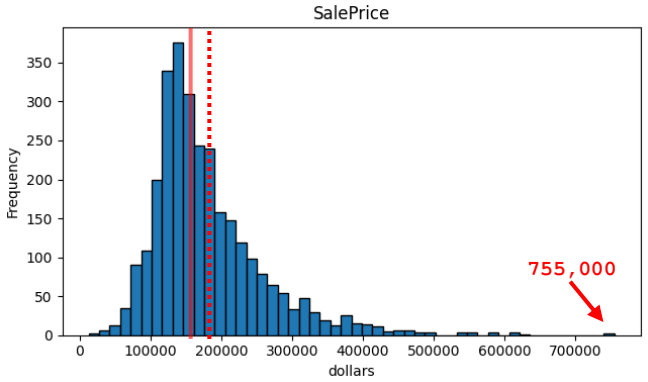
- *Alternatives to SCRIPS-DM Process Model (KDD and SEMMA)*
- *Our Working Environment*
- *Understanding Data (Data Attribution)*
- *EDA: finding problems, not making art*
- *Hands-on blocks in Colab*

Today's class:

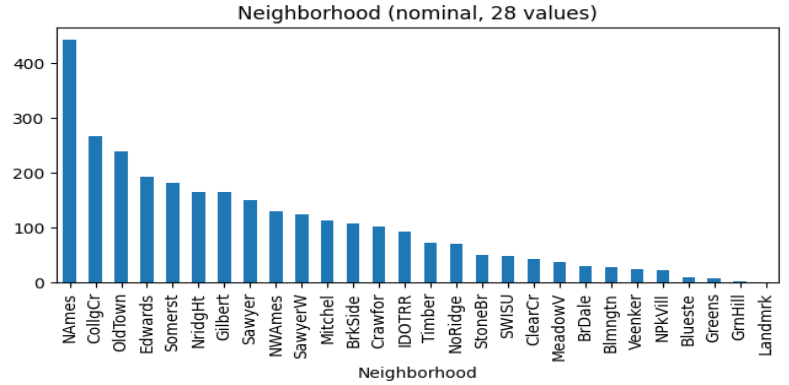
- ***Preprocessing: the eight techniques***
  - ***Data cleaning***
  - ***Discretization***
  - ***Binarization***
  - ***Attribute transformation***
  - ***Aggregation***
  - ***Sampling***
- ***Similarity and distance***
- ***The curse of dimensionality***

Where We Left Off Last Class?

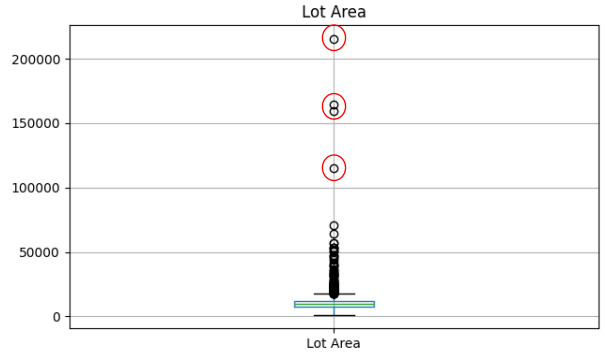
Problems in dataset (Ames Housing Dataset)



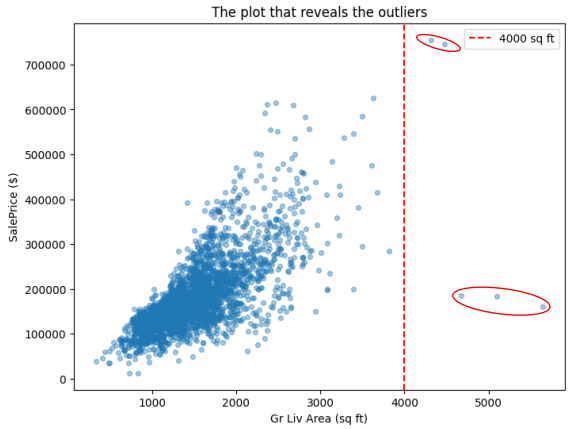
- Skewed right, so the mean sits above the median
- Max is 755,000, Median 160,000



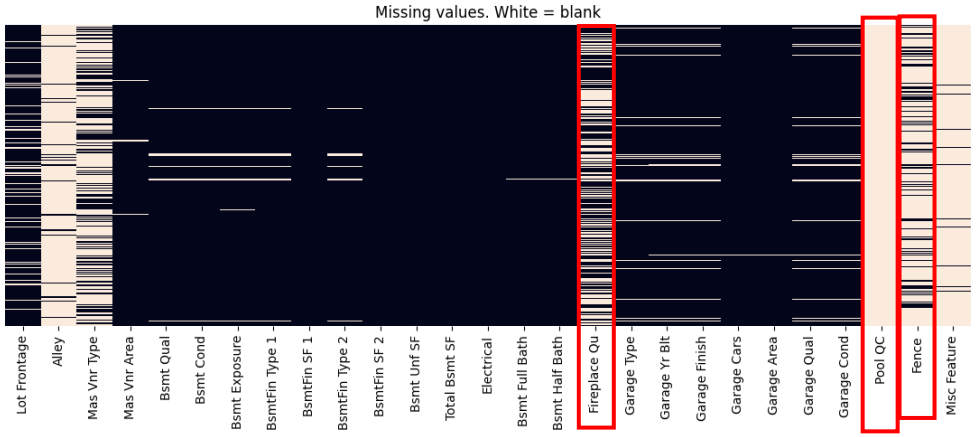
- 28 categories. Text, not numbers.
- One column becomes 28.



- 4 outliers. Real, not errors.
- Problem is scale, not cleaning.



- 5 outliers past 4,000 sq ft. 3 are partial sales.
- Problem is noise, not scale.



- Solid stripes: not applicable. Scattered gaps: not collected.
- Problem is missing values, and each kind needs a different fix.

Preprocessing

Definition:

Data preprocessing is the work of turning raw data into data an algorithm can actually use.

OR

Preprocessing is everything you do between loading the data and running the algorithm.

Three goals to achieve:

- **Make the data CORRECT:**
  Missing values, duplicates, outliers, noise.
- **Make the data USABLE:**
  Discretization, binarization, attribute transformation.
- **Make the data SMALLER:**
  Aggregation, sampling, dimensionality reduction, feature selection.

Preprocessing

Eight Techniques for the Three Goals

Technique	Goal	What it does	Covered
Data Cleaning	Correctness	Fix missing values, duplicates, outliers, noise	Today
Discretization	Usable	Continuous column into ordered bins	Today
Binarization	Usable	Anything into 0/1 columns	Today
Attribute transformation	Usable	Rescale or reshape the values	Today
Aggregation	Smaller	Combine attributes or objects into one	Today
Sampling	Smaller	Work with a representative subset	Today
Dimensionality reduction	Smaller	Fewer columns, keep the information	Sep 09
Feature subset selection	Smaller	Drop the columns that do not help	Sep 09

Section 2.3. Today is the first six.

Introduction to Data Mining, 2nd Edition

Preprocessing

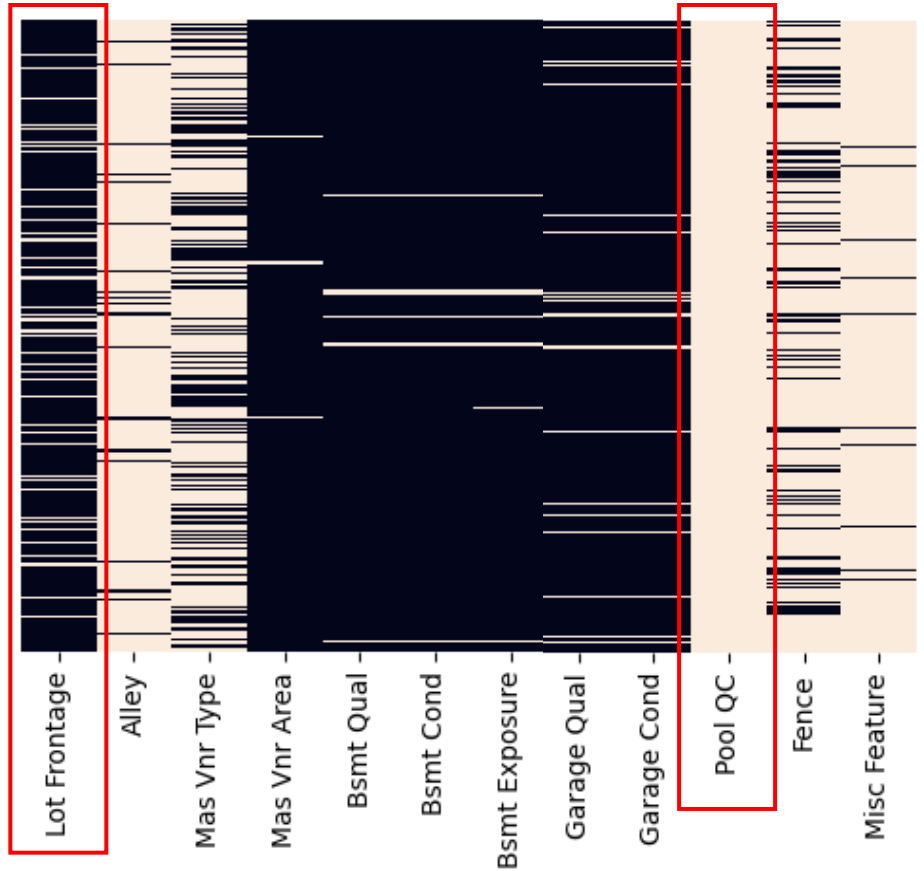
1. Data Cleaning (Missing Values: Two Reasons)

Information was not collected.

- People decline to give their age or their weight.
- In Ames: nobody recorded Lot Frontage for 490 houses.

The attribute does not apply.

- Annual income is not applicable to a child.
- In Ames: Pool QC is blank because the house has no pool.



Preprocessing

1. Data Cleaning (Missing Values: Three Responses)

Eliminate the object or the variable.

Cheapest. Also the most destructive. Check what it costs first.

- Drop the column when almost every value is blank. Pool QC is 99.7% empty.
- Drop the row when only a handful are affected. Electrical is blank for 1 house.

Estimate the missing value

Mean, median, mode, or a value from a neighbouring record.

- Time series: missing 2pm power reading, average the 1pm and 3pm values.
- Cyclical data: missing Tuesday 2pm, use last Tuesday 2pm instead.
- Grouped data: missing Lot Frontage, use the median for that Neighborhood.

Ignore it during the analysis

Some algorithms handle blanks natively. You do not always have to fix it.

- Decision trees and LightGBM learn a rule for absent values.
- df.mean() and df.corr() already skip blanks by default.

Preprocessing

1. Data Cleaning (Duplicates, Outliers, Noise)

- **Duplicates: the same record entered twice.**
  - Usually from merging data from different sources.
  - Inflates every count and every average.
  - Ames: 0 exact duplicates. Check anyway, always.

- **Outliers: a real value far away from the rest.**
  - The record is correct. The value is just unusual.
  - Ames: 3 lots above 100,000 sq ft. All real houses, zoned RL.

- **Noise: a wrong value.**
  - A typo, a broken sensor, a unit mix-up.
  - Ames: Garage Yr Blt contains 2207. No garage was built in 2207.

Outliers and noise look identical in a plot. Only domain knowledge tells them apart.

Preprocessing

1. Data Cleaning (Solving: Duplicates, Outliers, Noise)

- **Duplicates:**
  Decide what "the same" means before you delete anything.
  - Exact duplicate rows: drop them.
  - Same PID, different sale date: not a duplicate. The house sold twice.
  - Same person, two email addresses: needs fuzzy matching, not drop_duplicates().

- **Outliers:**
  Do not delete just because a boxplot flagged it.
  - Keep them, and scale or log transform so they stop dominating.
  - Remove them only when you can justify it. De Cock says drop the 3 partial sales.
  - Sometimes they are the whole point. Fraud detection looks for exactly these.

- **Noise:**
  Find it with a rule, not with a plot.
  - Range check: a year of 2207 is impossible. Flag it.
  - Cross-column check: Garage Yr Blt earlier than Year Built makes no sense.
  - Then treat the bad value as missing, and use one of the three responses from the last slide.

Preprocessing

2. Discretization (Turning Continuous Into Ordinal)

Turn a numeric column into a small number of ordered bins:

- Year Built has 138 distinct values. Becomes 4: Pre-war, Post-war, Modern, New.
- You lose precision. You gain readability and robustness.

Why anyone does this

- Some algorithms require it. Association rule mining cannot read continuous columns.
- Some algorithms work better with it. Decision trees, naive Bayes.
- Results become sentences people understand.

Bin	Years	Meaning
Pre-war	before 1940	Old construction
Post-war	1940 to 1979	Mid century
Modern	1980 to 1999	Recent, pre-2000
New	2000 and later	Newly built

Preprocessing

2. Discretization (Turning Continuous Into Ordinal)

Three Ways to Cut the Bins

Method	Where the cuts go	The problem with it
Equal width	Same interval size for every bin	Bins can end up nearly empty
Equal frequency	Same number of rows in every bin	Bin widths become very uneven
K-means	Where the data naturally clusters	Slowest, but usually the best cuts

Equal width is the trap.
On Sale_Price, three bins hold everything and one bin holds five mansions.

Preprocessing

3. Binarization (Everything Becomes Zero or One)

Turn a column into one or more 0/1 columns.

- A model does not read “Pconc” or “Cblock”. It reads numbers.
- Foundation column has 6 values. Becomes 6 columns of 0s and 1s.
- Each row has a 1 in exactly one column, 0 in the rest.

PConc = poured concrete
Cblock = cinder block
BrkTil = brick and tile
Slab = concrete slab
Stone
Wood

Foundation
PConc
BrkTil
Cblock
Wood



PConc	Cblock	BrkTil	Slab	Stone	Wood
1	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	0	0	0	0	1

Where you use it

- Any categorical column feeding a model that needs numbers. Regression, k-NN, neural networks.
- Association rule mining. Apriori only works on 0/1 data.
- Presence and absence questions: HasPool, HasGarage, WasRemodeled.

Preprocessing

3. Binarization (Everything Becomes Zero or One)

Other examples from Ames

- Central Air: Y or N becomes 1 or 0. One column, done.
- Pool QC: 99.7% blank becomes HasPool, a single 0/1 column.
- Neighborhood: 28 values becomes 28 columns.
- pandas calls it get_dummies. scikit-learn calls it OneHotEncoder

why not just number them, PConc=1, CBlock=2, BrkTil=3?

PConc = 1
Cblock = 2
BrkTil = 3
Slab = 4
Stone = 5
Wood = 6

Foundation
PConc
BrkTil
Cblock
Wood



Foundation
1
3
2
6

Because it creates a fake order.

Preprocessing

3. Binarization (Everything Becomes Zero or One)

Nominal and ordinal are not encoded the same way

- **Nominal: one-hot.**
  Foundation becomes 6 separate 0/1 columns. No order is implied.

- **Ordinal: map to ordered integers.**
  Kitchen Qual: Po=1, Fa=2, TA=3, Gd=4, Ex=5. The order is the information.

- Get this backwards and you break the data.
- One-hot an ordinal column and you throw the ranking away.
- Integer-encode a nominal column and you invent an order that is not there.

Preprocessing

4. Attribute Transformation (Change the Values, Keep the Meaning)

Apply a function to every value in a column

- The data does not change. The ruler changes.
- Two kinds: reshape the distribution or rescale the range.

Reshape: fix the skew

- log(x) squashes big values together, spreads small ones apart.
- Ames: SalePrice max is 755,000, median is 160,000. Long right tail.
- After log, the distribution is roughly symmetric. The mean means something again.
- Also works on Lot Area, where 3 lots flatten every other house.

Rescale: fix the range

- Standardization: subtract the mean, divide by the standard deviation.
- Ames: Lot Area runs to 215,000. Overall Qual runs to 10.
- After scaling, both count the same in a distance calculation.

Preprocessing

5. Aggregation (Combine Two Things Into One)

- Combine columns: several attributes become a single attribute.
- Combine rows: several records become a single record.

Three purposes:

- **Data reduction.** Fewer columns, fewer rows, less to process.
- **Change of scale.** Cities into regions. Days into months. Houses into neighborhoods.
- **More stable numbers.** Aggregated data varies less than the parts it came from.

Stability example:

- Monthly rainfall at one location swings wildly year to year.
- Yearly rainfall at the same location barely moves.
- Averaging cancels the random noise. The signal survives.

Preprocessing

5. Aggregation (Combine Two Things Into One)

Combining columns:

- Ames has 4 porch columns: Open Porch SF, Enclosed Porch, 3Ssn Porch, Screen Porch.
- Most are 0 for most houses. Individually, each is nearly useless.
- Add them into one Total Porch SF. Four columns become one, and it is a stronger feature.
- Same for bathrooms: Full Bath, Half Bath, Bsmt Full Bath, Bsmt Half Bath become Total Baths.

Combining rows:

- 2,930 individual house sales.
- Group by Neighborhood and take the median price. 2,930 rows become 28.
- Now you can answer "which neighborhoods are expensive?" in one look.

What you gain and what you lose:

- Gain: fewer columns, less noise, a clearer picture.
- Lose: you can no longer ask “does an enclosed porch add more value than a screen porch?”

Ask what question you are giving up before you aggregate.

Preprocessing

6. Sampling (Working With Less On Purpose)

- Statisticians sample because collecting everything costs too much.
- We sample because processing everything takes too long.
- Same technique. Different reason.

The principle:

- **Random without replacement.** Each row picked is removed. Nothing appears twice.
- **Random with replacement.** Rows stay in the pool. The same row can be picked again.
- **Stratified.** Split into groups first, then sample from each group.

Where Sampling Is Actually Needed

- Ames does not need it. 2,930 rows and 82 columns fits in RAM easily.
- Sampling matters when the data will not fit, or the algorithm is too slow to run on all of it.
- Instacart orders (3M+), Criteo click logs (4 Billion)

Similarity and Distance

Why We Need This

A realtor asks a question you cannot answer yet:

- "This house is going on the market. What are the comparable sales?"
- Comparable means similar. Similar to what, measured how?

The same question, four different ways:

- Which houses are most like this one?
  comparables
- Which houses group together naturally?
  clustering
- What did similar houses sell for?
  prediction
- Which house is like nothing else?
  anomaly detection

All four Questions need one thing: a number that says how alike two houses are.

Similarity and Distance

What a Distance Actually Is

Take two houses. Compare them column by column. Combine into one number.

- Small number means alike.
- Zero means identical.
- Big number means different.

Three words you will see

- **Similarity**: goes UP when things are alike. Usually 0 to 1.
- **Dissimilarity**: goes DOWN when things are alike. Starts at 0.
- **Distance**: a kind of dissimilarity. This is what we compute.

Two properties that always hold

- A house is zero distance from itself.
- Distance from A to B equals distance from B to A.

There is no single correct distance. There are several, and they disagree.
Choosing one is a decision you have to defend.

Similarity and Distance

Euclidean Distance

Euclidean distance is the straight-line length between two points in a space, much like a string stretched tight between them

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

Why square?

- It removes the minus signs. Distance is never negative.
- It makes big differences count much more than small ones.
- A difference of 10 contributes 100. A difference of 2 contributes 4.

It works in any number of columns

- 2 columns: this is Pythagoras. The straight line you would measure with a ruler.
- 82 columns: same formula, 82 terms.

Similarity and Distance

Example From Ames Dataset (Three Houses)

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

House	Lot Area	Overall Qual	Overall Cond	Gr Liv Area
A	5,280	4	7	1,361
B	8,712	4	7	952
C	4,800	4	5	1,844

A to B
Differences: −3,432 · 0 · 0 · 409
Squared: 11,778,624 · 0 · 0 · 167,281
Sum: 11,945,905
square root: **3,456.28**

A to C
Differences: 480 · 0 · 2 · −483
Squared: 230,400 · 0 · 4 · 233,289
Sum: 463,693
square root: **680.95**

So C is the comparable. 680.95 is much smaller than 3,456.28.

In A to B, Lot Area contributed 98.6% of the distance.
Overall Cond contributed 0.000%.

Similarity and Distance

Manhattan Distance:

Manhattan distance measures the distance between two points by summing the absolute differences of their coordinates, moving only in horizontal and vertical grid directions.

$$d(x, y) = |x_1 - y_1| + |x_2 - y_2| + \cdots + |x_n - y_n|$$

How it differs from Euclidean

- Euclidean squares each difference, so large gaps count disproportionately.
- Manhattan adds them directly, so every column contributes in proportion.

A to B: 3,432 + 0 + 0 + 409 = **3,841**
A to C: 480 + 0 + 2 + 483 = **965**

Same ranking. C is still nearest to A.

No squaring, but Lot Area still dominates. Manhattan does not fix the scale problem.

Similarity and Distance

Standardized Distances:

$$z = \frac{x - \text{mean}}{\text{standard deviation}}$$

Column	Mean	Std Dev
Lot Area	10,148	7,880
Overall Qual	6.09	1.41
Overall Cond	5.56	1.11
Gr Liv Area	1,500	506

Standardized values

House	Lot Area	Overall Qual	Overall Cond	Gr Liv Area
A	−0.62	−1.48	1.30	−0.27
B	−0.18	−1.48	1.30	−1.08
C	−0.68	−1.48	−0.50	0.68

Similarity and Distance

Standardized Distances:

Euclidean Distance After Standardization:

A to B → sum 0.843 → **0.92** **A to C** → sum 4.161 → **2.04**

Nearest to A is now B.

The comparison table for your final slide

	A to B	**A to C**	**Nearest**
Euclidean, raw	3,456.28	680.95	C
Manhattan, raw	3,841	965	C
Euclidean, scaled	0.92	2.04	B

A 3,432 sq ft difference in lot size is ordinary. A 2 point difference in condition is not. Raw numbers hid that completely.

The Curse of Dimensionality

Your 82 Columns Became ~300

- Neighborhood had 28 values. One-hot encoding made it 28 columns.
- Exterior 1st made 16. Foundation made 6. Roof Style made 6.
- 43 text columns encoded. 82 columns became roughly 300.

Nothing errored. It looked like a formatting step:

Two things broke:

1) **Sparsity:**
   - Volume grows exponentially with columns. Your rows spread thinner.
   - 2,930 houses cannot fill a 300 dimensional space. Nothing can.

2) **Distance concentration:**
   - Take 500 random points. Measure the closest pair and the farthest pair.
   - As columns increase, that gap shrinks toward nothing.